BME351- FLUID MECHANICS LAB

Course Outcome (CO)		Bloom's Knowledge Level (KL)
	At the end of course , the student will be able to	
CO 1	Understand the principles and performance characteristics of flow and thermal devices.	**K2**
CO 2	Know about the measurement of the fluid properties	**K1**
CO 3	Understand and analyze various properties of fluids	**K3**
CO 4	Evaluate the performance characteristics of fluid/thermal machinery	**K3**
CO 5	Evaluate the velocity and pressure variations in various types of simple flows.	**K3**
	DETAILED SYLLABUS	**0-0-2**

List of Experiments:(At least 8 of the following)

1. To determine the coefficient of impact for vanes.
2. To determine coefficient of discharge of an orifice meter.
3. To determine the coefficient of discharge of Notch (V and Rectangular types).
4. To determine the friction factor for the pipes.
5. To determine the coefficient of discharge of venturi meter.
6. To determine the coefficient of discharge, contraction & velocity of an orifice.
7. To verify the Bernoulli's Theorem.
8. To find critical Reynolds number for a pipe flow.
9. To determine the meta-centric height of a floating body.
10. To determine the minor losses due to sudden enlargement, sudden contraction and bends.
11. To show the velocity and pressure variation with radius in a forced vortex flow.

BME352- MATERIALS TESTING LAB

CO-1: Students will be able to perform different destructive and non-destructive testing methods to measure various mechanical properties. **(K2)**
CO-2: Students will be able to analyse the effect of different heat-treatment processes on the Hardness. **(K4)**
CO-3: Students will be able to simulate the material using simulating software / measure the mechanical properties of 3-D printed components. **(K3)**

List of Experiments: (At least 8 of the following)

1. Strength test of a given mild steel specimen on UTM with full details and stress versus strain plot on the machine.
2. Other tests such as shear, bend tests on UTM.
3. Impact test on impact testing machine like Charpy, Izod or both.
4. Hardness test of given specimen using Rockwell and Vickers/Brinell testing machines (this test should be conducted on MS Specimen – prior and after the various heat-treatment processes like Annealing / Normalizing / Hardening etc.; with this, 2 experiments can be designed: one for Annealing or Normalizing and another for Hardening / Tempering).
5. Fatigue test on fatigue testing machine.
6. Creep test on creep testing machine.
7. Study of NDT (non-destructive testing) methods like magnetic flaw detector, ultrasonic flaw detector, eddy current testing machine, dye penetrant tests.
8. Modelling of Graphene / any other materials structure using LAMMPs (LAMMPS Molecular Dynamics Simulator) or any other simulation software and studying the mechanical properties.
9. Measurement of mechanical properties (tensile test, compression test, hardness test / flexural test) of specimen made by 3-D printing

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